

Kiran's Project-Specific Real-Time Chat Application Handbook

Based on your actual project files

Project Overview

Real-time chat application using React.js, Node.js, Express.js, MySQL, JWT, bcrypt and Socket.io.

server.js

Creates Express server, Socket.io server, JWT middleware, online users tracking, private messaging, typing indicator, chat history loading and disconnect handling.

JWT Middleware

Reads token from socket.handshake.auth.token, verifies using jwt.verify(), stores decoded user in socket.user.

Online Users

onlineUsers object stores socketId and username. Broadcasted with io.emit('onlineUsers').

Private Messaging

Message saved to MySQL first, then emitted to receiver room and sender.

Typing Indicator

Frontend emits typing event, backend forwards it to receiver room.

Chat History

loadMessages query joins messages and users table and orders by created_at.

App.js

Handles socket events, login state, online users, chat selection, typing indicator, auto-scroll and message rendering.

React Hooks

useState, useEffect and useRef are used throughout the application.

Login.js

Authenticates users and stores JWT token in localStorage.

Register.js

Registers new users via backend API.

auth.js

Uses bcrypt.hash(), bcrypt.compare() and jwt.sign().

db.js

Creates MySQL connection and exports it.

Common Bugs Solved

EADDRINUSE, receiverId null, conditional hook issue, socket authentication issues.

Interview Question

Why Socket.io? Because chat applications require real-time bidirectional communication.

2-Minute Explanation

Secure chat application with authentication, online users, typing indicators and persistent message history.

Project-Specific Viva Questions

- Explain JWT middleware in your project.
- How does socket authentication work?
- How are online users tracked?
- How does private messaging work?
- How is chat history loaded?
- How does typing indicator work?
- Why use bcrypt?
- Why use Socket.io rooms?
- What bugs did you face and how did you solve them?
- How does useEffect work in your App.js?